

AREAS OF INTEREST

Machine Learning

Computer Vision

Machine Reasoning

Edge Computing

EDUCATION

•Direct Ph.D. in Electrical and Computer Engineering

2024–Present

Johns Hopkins University, Baltimore, MD

GPA: 3.93/4

– M.S. in Electrical and Computer Engineering earned in 2025

•B.Sc. in Electrical Engineering, Minor in Computer Science

2019–2024

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

GPA: 17.3/20 (3.75/4)

– Minor GPA: 17.5/20 (3.5/4)

•High School Diploma in Mathematics and Physics

2016–2019

Sampad High School, affiliated with NODET

GPA: 19.02/20 (4.0/4.0)

AWARDS AND HONORS

- **CVPR 2026 Broadening Participation Scholarship**, Received a \$600 travel stipend and registration waiver to attend CVPR 2026.
- **NeurIPS 2025 Student Volunteer**, San Diego; Selected student volunteer; received complimentary registration.
- **Second Best Paper Award**, AAAI 2025 Spring Symposium; For “GenAI at the Edge: Comprehensive Survey on Empowering Edge Devices.”
- **Best Poster Award (People’s Choice)**, Women in AI Symposium by Data Science and AI Institute (DSAI), Johns Hopkins University; For “MambaLiteSR: Image Super-Resolution with Low-Rank Mamba using Knowledge Distillation.”
- **DAC 2025 Young Student Fellow and Emerging Contributor Award**; Recognized at the 62nd Design Automation Conference, and received a \$500 travel stipend and registration waiver to attend DAC 2025.
- **Code-Cup Programming Competition**; Ranked 32/1073 in Data Science and 68/2417 in Algorithms.
- **National Universities Entrance Exam (Iran)**; Ranked 225/164,000 in Mathematics and Physics.

PUBLICATIONS

- **Class Unlearning via Depth-Aware Removal of Forget-Specific Directions** 2026
Arman Hatami, Romina Aalishah, Ilya E. Monosov
– CVPR 2026; Oral Presentation at the Workshop on Machine Unlearning for Computer Vision
- **MedMambaLite-v2: Shared Selective Scan for Efficient Edge Medical Mamba** 2026
Romina Aalishah, Mozghan Navardi*, Nithin Kidangazhiath Mana, Tinoosh Mohsenin*
– IEEE Transactions on Biomedical Circuits and Systems (TBioCAS), Special Issue: Selected Papers from BioCAS 2025. Issue of Aug 2026.
- **EdgeNavMamba: Mamba-Optimized Object Detection for Energy-Efficient Edge Devices** 2025
Romina Aalishah, Mozghan Navardi, Tinoosh Mohsenin
– 11th IEEE International Conference on Edge Computing and Scalable Cloud (IEEE EdgeCom 2025)
- **MedMambaLite: Hardware-Aware Mamba for Medical Image Classification** 2025
Romina Aalishah, Mozghan Navardi, Tinoosh Mohsenin
– 21st IEEE Biomedical Circuits and Systems Conference (IEEE BioCAS 2025)
– Invited to submit an extended version of this paper to the Special Issue of selected papers from BioCAS 2025 at the IEEE Transactions on Biomedical Circuits and Systems journal.
- **GenAI at the Edge: Comprehensive Survey on Empowering Edge Devices** 2025
Mozghan Navardi, Romina Aalishah, et al.
– AAAI Spring Symposium 2025; Special Session: Generative AI @ Edge
– Won the Second Best Paper Award at this session.
- **MambaLiteSR: Image Super-Resolution with Low-Rank Mamba using Knowledge Distillation** 2025
Romina Aalishah, Mozghan Navardi, Tinoosh Mohsenin
– ISQED Spring Symposium 2025; Special Session: Generative AI in Edge

RESEARCH EXPERIENCES

•Research Assistant

Johns Hopkins University, LAMI

Feb. 2026 - Present

Baltimore, MD

- Working at the Laboratory of Adaptive and Maladaptive Intelligence (LAMI) under [Prof. Ilya Monosov](#).
- Analyzing interpretability in deep neural networks by studying how internal representations drive model behavior.
- Using knowledge distillation as a controlled framework to compare representations across two networks solving the same task and quantify how compression changes internal computation.
- Evaluating how representation differences between teacher and student predict failure modes and using these insights to guide more interpretable and robust models.

•Research Assistant

Johns Hopkins University, EEHPC Lab

Sep. 2024 - Jan. 2026

Baltimore, MD

- Working at Energy-Efficient High-Performance Computing (EEHPC) laboratory under [Prof. Tinoosh Mohsenin](#).
- Contributing to energy-efficient computing projects focusing on computer vision applications.
- Implementing hardware-aware optimization for compressed deep neural networks to enhance deployment on resource-constrained devices.
- Developing vision-language-action models in robotics projects focused on path planning by leveraging reinforcement learning and brain-inspired deep learning algorithms to enhance autonomous navigation.
- Using IsaacLab, MiniWorld, and Gazebo simulator for better assessment and decreasing the sim-to-real gap.

•Teaching Assistant

Johns Hopkins University

Aug. 2025 - Dec. 2025

Baltimore, MD

- Working as the TA for the course "Design of Advanced Instruments and Systems" under [Mr. Sathappan Ramesh](#).

•Research Assistant

Amirkabir University of Technology(Tehran Polytechnic)

Mar. 2024 - Jul. 2024

Tehran, Iran

- Working at the Laboratory of Computational Intelligence under [Prof. Mohammad Bagher Menhaj](#) and [Prof. Masoud Shafiee](#).
- Using reinforcement learning on mobile robots to discover an area.

•Signal Processing Intern

West Pomeranian University of Technology, Faculty of Electrical Engineering

Mar. 2023 - Jun. 2023

Szczecin, Poland

- Work under [Prof. Grzegorz Psuj](#).
- Reviewing resources and comparing the methods related to Magnetic Barkhausen Noise, data processing and knowledge extraction, and deep learning-based robust analysis of magnetic Barkhausen noise using Matlab.

•Back-End Developer

GoldScan

Feb. 2022 - Dec. 2023

Tehran, Iran

- Developing back-end using Django, Django-Rest framework
- Utilizing such technologies as HTML, CSS, and SQL to develop new applications
- Using Postman to test these applications
- Creating Python-Telegram Bot for some parts of projects.

•Electrical Engineering Intern

SYNTECH Technology and Innovation Center at Qazvin Islamic Azad University

Sep. 2021 - Mar. 2022

Qazvin, Iran

- Interface Electronic Circuits to Arduino
- VHDL coding
- Read and analyze papers in the field of Biomedical Engineering (Related to the department projects)

PROJECTS

•Analysis and Assemblage of Comau Smart 5 NJ-165-3.0 Robot

Amirkabir University of Technology

May 2022 - Jun. 2022

- Analysis, simulation, design, programming, 3D printing, and assemblage of the robot with a scale of 1:15 of the main size.
- Technology Used: SOLIDWORKS, MATLAB, Simulink, Arduino.

•Audit Platform

GoldScan

Dec. 2022 - May 2023

- Provide the back-end side of an audit platform that automates the auditing process for ledgers, calculating the risk of auditing based on the ledger data and generating worksheets that can be used by auditors to perform the audit.
- Data processing of a given ledger data.
- Provide unit tests to check the performance and use Postman to test the functionality.
- Refactor some parts for better performance.
- Technology Used: Python, Django, Django REST Framework, PostgreSQL, Docker.

LANGUAGES

English: Advanced proficiency, TOEFL (iBT) score: 105/120

Persian: Native proficiency

German: Conversational proficiency

EXTRACURRICULAR ACTIVITIES

•CISS Staff

Mar. 2025

Johns Hopkins University

Baltimore, MD

- The Conference on Information Sciences and Systems (CISS) is a forum for scientists, engineers, and academics to present their latest research results and developments in multiple areas of Information Sciences and Systems.
- Contributing to the registrations, preparations, and organizing of the conference.

•Chess Team Member

Aug. 2022 - Mar. 2024

Amirkabir University of Technology (Tehran Polytechnic)

Tehran, Iran

- One of the 4 members of university chess team
- The team ranked 5th in the National Universities Rapid Chess Championship 2022.
- The team ranked 7th in the National Universities Blitz Chess Championship 2022.

•Local Committee Member

Jan. 2022 - Feb. 2023

IAESTE Iran

Tehran, Iran

- IAESTE is an international organization exchanging students for technical work experience abroad.
- Socialize with interns from other countries and guide them.